Given

fc = 1 kHz

C = 10 nF

$$R = \frac{1}{2\pi f_c C}$$

$$= \frac{1}{2\pi(1000)(10\times10^{-9})}$$

$$= 15.9k\Omega$$

Doubling R/C

fc = 500 Hz
(R=31.8kohm or C=20nF)

Halving C/R

fc = 2 kHz
(R=7.95kohm or C=5nF)